

# Explain code steps

|                                          |   |
|------------------------------------------|---|
| 1. Vector.cppFileexplain the code. ....  | 2 |
| 2. is there a for each loop in stl. .... | 6 |



## 2. is there a for each loop in stl

Yes, C++ STL (Standard Template Library) supports a **for-each loop**, introduced in **C++11**, using the **range-based for loop** syntax.

### ✓ Syntax:

```
cpp

for (auto element : container) {
    // use element
}
```

### ✓ Example using vector<int>:

```
cpp

vector<int> v = {1, 2, 3, 4, 5};

for (auto x : v) {
    cout << x << " ";
}
```

This will output:

```
1 2 3 4 5
```

### 🔄 Modify Elements using Reference:

If you want to **modify elements**, use reference (&):

```
cpp

for (auto &x : v) {
    x *= 2;
}
```

Now **v** becomes {2, 4, 6, 8, 10}.

## Under the Hood:

This loop uses the `begin()` and `end()` iterators of the container, just like this:

```
cpp

for (auto it = v.begin(); it != v.end(); ++it) {
    // same as range-based for
}
```

## Common Mistake:

```
cpp

for (auto x : v) {
    x *= 2; // Doesn't change original vector
}
```

This creates a copy (x), so changes won't reflect on the actual vector.

Let me know if you want a version of your original code rewritten using for-each loops!